

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0884

Beardsley St 138 kV
June 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0884. The Interconnection Customer proposes a 169.4 MW wind facility interconnecting at the Beardsley St 138 kV substation in the Virginia area. The request was received on October 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 305 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,696,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Salem Electric Department-Neosho 138.0 kV	Montello-Neosho 138.0 kV line	138.3	165	41.5
Beardsley St-Neosho 138.0 kV	Lewistown No. 1-Beardsley St 138.0 kV Line	133.3	249	62.5
Hancock-Ash St 138.0 kV	Beardsley St-Lewistown No. 1 138.0 kV	125.4	232	62.5
Hondo Substation-Lewistown No. 1 138.0 kV	Beardsley St-Neosho 138.0 kV L	122.2	185	47.2
Lewistown No. 1-Beardsley St 138.0 kV line	Berdsley St-Neosho 138.0 kV Line	116.3	212	47.2
Ash Street-Montello 138.0 kV	Salem Electric Department-Neosho 138.0 kV	110.1	203	41.5
Montello-Neosho 138.0 kV Line	Neosho-Salem Electric Department 138.0 kV line	108.7	206	41.5
Leesville-Clover 345.0 kV Line	Riverview Substation-Person 345.0 kV	100.7	1,241	23.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Salem Electric Department-Neosho 138.0 kV	Rebuild line with higher-capacity conductor	\$948,000	96% (\$908,000)
Beardsley St-Neosho 138.0 kV	Rebuild line with higher-capacity conductor	\$6,399,000	77% (\$4,912,000)
Hancock-Ash St 138.0 kV	Rebuild line with higher-capacity conductor	\$3,402,000	77% (\$2,612,000)
Hondo Substation-Lewistown No. 1 138.0 kV	Rebuild line with higher-capacity conductor	\$4,098,000	78% (\$3,218,000)
Beardsley St Substation-Lewistown No. 1 Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$4,419,000	78% (\$3,470,000)
Ash Street-Montello 138.0 kV	Rebuild line with higher-capacity conductor	\$2,042,000	96% (\$1,954,000)
Montello-Neoso 138.0 kV	Reconductor line	\$374,000	96% (\$358,000)
Leesville-Clover 345.0 kV Line	Reconductor line	\$69,923,000	6% (\$4,264,000)
	Total allocated to project		\$21,696,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.